Mark K Cowan

Engineering Operator

Tallinn, Estonia

hackology.co.uk

mark@battlesnake.co.uk

github.com/battlesnake

linkedin.com/in/photonicist

Skills

- C++
- C
- Python
- Bash
- Docker
- Analog electronics
- Digital electronics
- Embedded systems
- Java
- C#
- JavaScript, TypeScript, Node.js
- Golang
- Kotlin

Experience

2026– Founder, HiPPO — Potamus

Helping organisations to smoothly scale their engineering operations

In tech, ops is usually an afterthought until it becomes a noose on velocity.

I can help you to solve that, or to prevent it.

2024–2026 CTO, Co-founder — Lendurai

End-to-end FPV strike missions with terminal guidance

Most details confidential. Designed initiation board used for every live-fire FPV exercise in Estonia throughout 2024.

C++, ARM64 assembly, i.MX8, Docker, Qemu, OpenCV, PX4, MAVLink, LoRA, Bash, Slack, Github, ...

2023–2023 Occasional merc — Kanvas.ai Art Technologies

Kanvas.ai provides fractional investment in fine art, both via traditional means and also via tokenisation on the Tezos blockchain.

Next.js, AWS, Tezos

2019–2024 Senior Data Engineer — Bolt

Delivering new operations/tools/infra to scale from <10 models to >1000, from quarterly releases to daily/weekly releases, and improving performance and costs.

AWS (EC2+S3+Sagemaker+Redshift), Airflow, SQL, Python, Docker, Jenkins, Bash, TypeScript, ...

2018–2019 Lead flight software engineer — Open Cosmos

Designing the next-generation flight software (C/Kotlin).

C, C++, Kotlin, devicetree, Linux kernel, Das U-boot, Kicad, Java/Swing, ...

2017–2018 Head of software — Open Cosmos

Building and growing a team of skilled software engineers, aerospace engineers & physicists.

HipChat, Trello, Gitlab, ...

2015–2016 Software & payload engineer — Open Cosmos

Delivered flight-software for our first satellite, foundations for mission-control software, ground-segment radio software, ground-station networking, tracking, ...

C, FreeRTOS, AVR32, Node.js, Ettus USRP SDRs, Docker, ZeroMQ, Typescript, Angular, Postgres, Java, Gradle, Das U-boot, Linux kernel, systemd, Kicad

2015–2015 Thundercloud Tech

Short-lived startup while on Entrepreneur First.

Improving the UK power grid's low-voltage network's efficiency and reliability.

2015–2015 Entrepreneur in residence — Entrepreneur First

Invited onto Entrepreneur First cohort #5, from where I saw the start of Open Cosmos, which I joined a few months later.

2014–2015 Front-end developer — Eesti Rahvusringhääling

The usual culprits: HTML5, CSS3, JavaScript. Also C#, .NET MVC4, Entity Framework, MS SQL.

2012–2012 Lab demonstrator — University of Manchester

Teaching and supervising laboratory projects for BSc Chemical Engineering students, involving real-time tomographic imaging of mixing processes.

2011–2011 TracerCo — TracerCo

☢, ⚠, and MODBUS.

2006–2011 Head coach / coach / assistant coach — *Various squash clubs*

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More skills

Squash

StarCraft II

Counter-strike

Beat Saber

Music tech

Ice-skating

Guitar

Curling

Piano

Sense of humour

Education

2013–2013 Machine Learning — Coursera

Andrew Ng's course on Coursera, 100% score.

2011–2013 PhD Chem. Eng. & Analytical Sci. — University of Manchester

I left the PhD during 2nd year. I taught a lab project involving real-time tomographic imaging of mixing processes.

I took several extra taught courses including:

- COMSOL Multiphysics
- NMR spectroscopy
- OpenCL
- OpenACC
- OpenMP
- MPI
- Intel Parallel Studio XE

2010–2011 MSc Photon Science — University of Manchester

Distinction in:

- Holography and imaging
- Optical instruments
- Soft-matter physics

Merit in:

- Semiconductor quantum structures,
- Laser technology
- Laser photomedicine
- Lasers and photonics
- Soft-matter physics

I was elected as representative for the course.

I was elected as treasurer for the post-graduate society.

2007–2010 BSc (Honours) Maths and Physics — University of Leeds

First class in:

- Advanced quantum mechanics
- Medical imaging
- Calculus and mathematical analysis
- Modelling with differential equations
- Intro to music technology

Upper second-class includes (non-exhaustive):

- Nuclear physics & energy
- Advanced mathematical methods
- Further linear algebra
- Project (distributed computer simulation)

I represented the university nationally at:

- Squash
- Shodokan Aikido
- Korfball
- Tenpin bowling

2000–2007 High school / Sixth form — Lancaster Royal Grammar School

Advanced (A) level in:

- Maths
- Further Maths
- Physics
- Chemistry
- Critical thinking

Various awards including:

- Community Sports Leadership award
- "Distinction" in British Informatics Olympiad
- Consistent "Gold" awards in several Mathematics Olympiads

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Want to learn

- Rust
- Golang
- Eesti keel
- FPGA + VHDL development
- Illustration
- VFX
- Some form of dance
- ASIC design & development
- High-speed+density PCB design

Miscellaneous

— **References** — LinkedIn

References are immediately available on my LinkedIn profile at https://www.linkedin.com/in/photonicist/.

See the "Recommendations" section on the profile.

— **Patent** — GB201603920A

Apparatus and method for satellite payload development.

2016 **Hackathon** — ActInSpace @ Tallinn

Won fast-track to Prototron accelerator with lunar agriculture concept.

2018 **"Boom" generative graphics/audio project** — JS+GLSL+Povray

A HTML5 canvas + WebGL + GLSL + WebAudio project, to experiment with procedural generation.

There's no real objective or meaning in this artwork, it's just for fun.

It'll never be finished, I'm always adding layers and sections to it when I get inspiration.

Live at https://hackology.co.uk/boom/.

Due to WebGL+WebAudio usage, this will not work on most smartphones/tablets. It should work on any half-decent PC running Chrome or Firefox.

2013 **"neural"** — CTAN/LaTeX

A LaTeX package for drawing neural-network diagrams with Tikz.

100+★ on Github.

2015 **"angular-chrome"** — Chrome/JS

A Chrome extension for inspecting AngularJS contexts.

Available on the Chrome Store, but the URL is huge.If your PDF viewer supports hyperlinks then you can just click here to open it.

It has ten reviews averaging slightly over 4.3/5★ at the time of writing this.

On Github at github.com/battlesnake/angular-chrome.

2015 **"gulp-google-webfonts"** — npm/node.js

A node.js/gulp package for automatically downloading fonts from Google WebFonts during the build process, and auto-generating the relevant CSS.

Consistently >1000 weekly downloads from npm at npmjs.com/package/gulp-google-webfonts.

50+★ on Github at github.com/battlesnake/gulp-google-webfonts.

Used by BitWarden project among others.

2015 **"kaiu"** — C++

Self-learning project: A collection of interacting asynchronous and concurrent experiments in C++.

Not used in production, nor should it as the heavy templated nature of this library means that compile times will explode massively.

On Github at github.com/battlesnake/kaiu.